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Manuscript Title: Validity of the Elite HRV Smart Phone Application for Examining Heart Rate Variability in a Field Based Setting

Running Head: Validation of smart phone HRV app in field setting

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ABSTRACT

The introduction of smart phone applications has allowed athletes and practitioners to record and store R-R intervals on smart phones for immediate heart rate variability (HRV) analysis. This user-friendly option should be validated in the effort to provide practitioners confidence when monitoring their athletes before implementing such equipment.

The objective of this investigation was to examine the relationship between a vagal related HRV index, rMSSD when derived from a smart phone application accessible with most smart phone operating systems against Kubios HRV 2.2

R-R intervals were recorded immediately upon awakening over 14 consecutive days using the Elite HRV smartphone application. R-R recordings were then exported into Kubios HRV 2.2 for analysis. The relationship between $rMSSD_{ln}$ derived from Elite HRV and Kubios HRV 2.2 was examined using a Pearson Product Moment Correlation and a Bland-Altman Plot.

An extremely large relationship was identified ($r = 0.92$; $p < 0.0001$; CI 95% = 0.90;0.93). A total of 6.4% of the residuals fell outside of the 1.96 ±SD (CI 95% = -12.0%; 7.0%) limits of agreement. A negative bias was observed (mean: -2.7%; CI 95% = -3.10%;-2.30%) whose CI 95% failed to fall within the line of equality.

Our observations suggest Elite HRV as a non-valid smart phone application for examining $rMSDD_{Ln}$ when compared to Kubios HRV 2.2. We propose further research is warranted to

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confirm our observations and to identify if this smart phone application may be reliable rather than valid when assessing parasympathetic modulation.

Key Words: Autonomic Modulation, R-R intervals, Monitoring, Athletes

INTRODUCTION

Heart rate variability (HRV) has become a popular, non-invasive, physiological assessment tool with practitioners for monitoring the activity of the autonomic nervous system (4, 26). Previous literature has demonstrated how practitioners can examine the magnitude of autonomic modulation through examining resting heart rate and the variability between beat to beat (R-R) intervals in response to a training stress (6, 26), psychological stress (7, 27), alterations in blood volume (5) and hormonal fluctuations (14, 16). It has been suggested that the very purpose of monitoring HRV is to develop insight into how an athlete adapts to a training stress during a training period or through competition (4). Insight into an athlete's autonomic nervous system response from a training stress may help integrative support staff and coaches adapt future training loads in an effort to facilitate improved recovery and increased performance potential (17, 18).

The non-invasive nature of collecting R-R intervals through a valid and personal heart rate (HR) monitor outside of a laboratory setting is thought to have enhanced the ability to collect resting heart rate recordings for immediate and long-term HRV monitoring (20). When monitoring HRV in athletes, current evidence suggests a short-term collection period ranging from 5-10 minutes, immediately upon awakening in the morning, while in a supine position (24-26). However, the necessary time required to individually export R-R files, perform artifact

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corrections, and analyze each athlete's HRV may often prevent timely feedback to coaching staff, preventing the opportunity to make adjustments in training load for proceeding on field sessions. Furthermore, integrative support staff may be limited to only accessing recorded R-R files prior to training when athletes tend to return HR monitors for data extraction. Such a circumstance with limited turnover time may impact the capability of practitioners to make decisions for imminent training sessions without any direction from HRV. Perhaps the most common, yet overlooked obstacle when working in a team setting is the failure to return HR monitors for data extraction. Such a situation can lead to athletes training without a HR monitor during a session whereby HR derived training loads become absent. Recent literature has proposed practitioners are often limited to athlete adherence and what measures are most practical when monitoring physiological variables such as HRV (4, 23). Current evidence towards smart phone applications (apps) capable of syncing with a personal HR monitor for collection R-R intervals have provided a user-friendly software for the instant analysis of HRV(3, 8). Furthermore, HRV smart phone apps often provide the option to export raw R-R data files directly to the practitioner's computer via email. The idea of team sport athletes possessing their own personal HR monitor, compatible with an HRV smart phone app capable of collecting R-R intervals, that can be exported to the practitioner prior to training would be a convenient and optimal experience for both athletes and coaching staff. The purpose of this investigation was to examine the relationship between a popular smart phone app, Elite HRV, and its validity to examine a vagal related HRV index as expressed through rMSSD when compared to Kubios HRV 2.2 (28) in a field setting. In conducting this investigation we hypothesized that the smartphone application, Elite HRV will be identified as a valid platform for examining rMSSD when compared to a commonly utilized HRV analysis software program Kubios HRV 2.2.

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METHODS

Experimental Approach to the Problem:

This is a validation study design where the relationship and level of agreement in $rMSSD_{ln}$ derived from a smart phone application and a validated computer software program was examined over a 14 day period. A Pearson Product Moment Correlation will identify the level of relationship along with a Bland Altman Plot demonstrating the level of agreement.

Subjects

A sample consisting of 29 male and 8 female recreational athletes acting as participants each took part in a 14 day field study in which testing occurred away from a laboratory setting. Characteristics are provided in Table 1. All participants were asked to provide informed written consent prior to the initiation of testing in accordance with guidelines of the Clinical Research Ethics Review Board at the University of British Columbia who approved collection of data for this study. All participants completed the Physical Activity Readiness Questionnaire for Everyone (30). Additional information regarding the participants' health history was also collected through administration of additional questionnaires during the 17 days.

TABLE 1 HERE

Procedures

Data collection occurred in a field-based setting where all participants lived in a dormitory living environment for the entire study period. Participants were provided a Polar H7 HR monitor (Polar Electro Oy, Kempele, Finland) with a sampling frequency of 1000 hz. Previous literature has demonstrated the use of a Polar H7 as a valid HR monitor when recording

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R-R intervals for HRV analysis (10). Participants were previously educated on how to operate their designated HR monitor and smart phone application (Elite HRV) prior to the start of data collection. Participants were instructed to record 10 minutes of resting heart rate, double the identified minimum time frame required for examining time domain indices (29); over 14 consecutive days immediately upon awakening in the supine position. The extended recording length was implemented as each participant was able to provide the time in the effort to capture additional cardiac cycles for analysis. A time domain indices, rMSSD (i.e. root mean square of successive R-R intervals) was selected to reflect alterations in vagal modulation and was preferred due to its enhanced reliability in demonstrating parasympathetic activity when compared to other power spectral density indices (1). Both breathing rate and depth were left uncontrolled for and participants were instructed to relax and breathe at a natural rate. Prior investigations have identified rMSSD to possess robust statistical properties capable of negating significant influence towards R-R interval length from respiratory sinus arrhythmia when not controlling for breathing rate and depth (22). All raw R-R data files were exported to a laptop computer at the completion of the 14 days using Elite HRV's email exporting option. All smart phone generated $rMSSD_{ln}$ values were recorded manually by the investigators each morning after collection during the entire study period.

All R-R files were exported from the Elite HRV smart phone application via email into a separate computer for analysis using Kubios HRV 2.2. Each file was corrected for ectopic beats and artifact prior to analysis using an artifact correction method provided in Kubios HRV 2.2. A medium level of artifact correction identifying R-R intervals varying above or below 0.25 seconds compared to the average was chosen to help preserve the variability while addressing the

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presence of any artifacts. Kubios HRV 2.2 utilizes a piecewise cubic spline interpolation method to generate corrupted or missing values.

Statistical Analyses

Raw rMSSD data was log transformed using the natural log rhythm for addressing any non-uniformity of error. All values as identified as outliers where included in the data set and its analysis. As previously stated in the procedures section, we believe the standardization throughout the data collection period, its filtering, logarithmic transformation and its analysis, would allow for the identification of any outliers to largely be due to biological error resulting from physiological perturbations in homeostasis which was the very purpose for examining HRV. Levels of agreement between Kubios HRV 2.2 and Elite HRV were assessed using a Bland Altman Plot (2), comparing the % residuals between each method to the average values. Reasonable agreement was defined as <5% of the residuals residing outside of 1.96 SD (CI 95%) from the mean. Bias was shown using a CI 95% and its position relative to the line of equality. A Pearson Product Moment Correlation expressed the strength of relationship between Elite HRV and Kubios HRV 2.2. A correlation coefficient of r >0.90 (CI 95%) was chosen to indicate an acceptable level of relationship (13). Statistical analysis was completed using Microsoft Office Excel 2010.

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RESULTS

rMSSD values derived from Elite HRV were non-normally distributed (Kurtosis = 1.3; Skewness = 1.1), and are expressed using a Q-Q plot seen in Figure 3. rMSSD values derived from Kubios HRV 2.2 were normally distributed. These observations prevented further analysis between non-log transformed rMSSD values from Kubios and Elite HRV. $rMSSD_{ln}$ values derived from Elite HRV demonstrated a normal distribution (Kurtosis: -0.6; Skewness: -0.3); a visual examination using a Q-Q plot can be seen in Figure 4.

FIGURE 4 HERE

When examining the $rMSSD_{ln}$ values produced through Kubios and Elite HRV, our results demonstrated an extremely large relationship ($r = 0.92$; $p < 0.0001$; CI 95% = 0.90-0.93) (11) as seen in Figure 1. A total of 6.4% of the residuals fell outside of the 1.96 ±SD (CI 95% = -12.0;7.0%) limits of agreement as observed in Figure 2. A negative bias was identified (mean: -2.7%; CI 95% = -3.10%;-2.30%) with a CI 95% failing to reside within the line of equality as displayed in Figure 3.

FIGURE 1 HERE

FIGURE 2 HERE

FIGURE 3 HERE

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DISCUSSION

The purpose of this investigation was to examine the validity of a HRV smart phone app commonly accessible to all smart phone operating systems for assessing autonomic modulation as expressed through a parasympathetic index, rMSSD, in a field based setting. The correlation we observed (r = 0.92; CI 95% = 0.90;0.93, $p < 0.0001$) demonstrated an extremely large relationship (11) between Elite HRV and Kubios HRV 2.2 and is in agreement with previous literature examining the validity of other smart phone applications (3, 8). However, when identifying a new measurement technique for the purpose of replacing an already valid technique, further statistical support beyond a strong correlation focusing on the limits of agreement should be implemented (2, 9). The results from the Bland Altman analysis demonstrated that 6.4% of the $rMSSD_{ln}$ residuals fell outside of the priori levels of agreement set at CI 95% (Figure 2). Furthermore, a negative bias was observed (mean: -2.7%; CI 95% = -3.10%;-2.30%) with its CI 95% failing to reside within the line of equality as seen in Figure 3. When visually examining all residuals, our results indicate larger discrepancies with a negative bias towards larger $rMSSD_{ln}$ values (Figure 2).

A potential explanation for the lack of agreement observed in this study compared to previous investigations examining HRV smart phone applications may have been due to the difference in R-R interval recording periods. Sampling periods for each participant varied beyond the predetermined 10 minutes as some participants lost track of time. The Elite HRV smart phone application utilized in this investigation did not permit specific selections or the partitioning of R-R intervals beyond that of the total recording period collected (ie. ≥10 minutes). When examining R-R recordings, longer sampling lengths have been shown to increase variance in R-R cycles and vagal related HRV indexes (29). For this reason it has been advised against

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comparing repeated measures in HRV that were derived from different sampling lengths (29). However, for the purpose of this investigation, all R-R sampling periods regardless of the recording length, were compared against themselves and such violations therefore would not exist. Alternatively, the differences in residuals observed may be indicative of discrepancies in the artifact correction techniques provided in Elite HRV and Kubios HRV 2.2. The large residuals observed in the current study may have been elicited from an extended sampling period beyond that of the predetermined 10 minute recording period. Such a prolonged period may have increased the potential for the collection of ectopic beats and artifacts (29). Furthermore, HRV analysis derived from shorter recording periods have indicated greater sensitivity to artifact correction techniques, potentially explaining the magnitude of residuals currently observed between short and long recording periods (29).

A medium level of artifact correction which identifies R-R intervals varying above or below 0.25 seconds compared to the average was chosen to help preserve the variability while addressing the presence of any artifacts when analyzing through Kubios HRV 2.2. Kubios HRV 2.2 utilizes a piecewise cubic spline interpolation method to generate corrupted or missing values, a preferred method for smoothing curves using a third degree polynomial (15)**.** To our knowledge, the technique and level of artifact correction implemented in Elite HRV is not made known. As such, a comparison between each technique of artifact removal was not possible. There remains a lack of standardized recommendations in the literature on how best to edit R-R recordings containing artifact. It has been suggested that most practitioners hold a minimum requirement in normal R-R intervals of 80% before editing artifacts for further analysis in HRV (21). Further investigations towards the artifact removal techniques from available smart phone

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applications are required in the effort to provide practitioner's confidence in the HRV values produced.

In conclusion we identified an extremely large correlation between $rMSSD_{ln}$ values elicited through Elite HRV and Kubios HRV 2.2. Contrary to our hypothesis, the limits of agreement were not met as 6.4% of the residuals fell outside of the predetermined priori set a CI 95% as well as recording a negative bias of -2.7% whose CI 95% failed to overlap the line of equality. It is also recommended that further investigations between HRV smartphone applications and various software programs disclose the technique and level of artifact correction utilized.

PRACTICAL APPLICATIONS

The use of a smart phone application for assessing HRV in athletes has become popular amongst integrative support staff, particularity in team sports. This option is often viewed as efficient and practical when monitoring athletes in a team setting as the app is capable of collecting, analyzing and exporting vagal related HRV indexes directly to the practitioner for further examination and recording. The ability to analyze, store and export HRV recordings may be most practical for non-centralized athletes and with teams travelling with minimal staff. This option would allow for the continuing physiological support from practitioners not traveling provided internet access is available for exporting data. In conjunction with the level of validity a sport science assessment tool offers, its ability to remain reliable may be equally as important as its validity. The ability of an instrument to reliably produce similar results when repeated multiple times should be of great important to the integrative support staff when desiring to

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monitor the long term development of their athletes. Current evidence demonstrate sport science instruments commonly elicit larger sampling errors as the value being measured increases (19), an issue that may become problematic when measuring large HRV values. However, the logarithmic transformation of the raw rMSSD values have been suggested to correct this non uniformity of error whereby applying a similar magnitude of error for all values measured (12). Although this investigation solely examined the validity of a popular smart phone HRV application, recent reviews have postulated that the occurrence of re-occurring errors after automatic filtering should not prevent practitioners from effectively monitoring their athletes when this filtering method is standardize for each analysis (4). While the present study demonstrated an unsatisfactory level of agreement between Kubios HRV 2.2 and Elite HRV, the relationship observed as expressed through the Pearson's Product Moment Correlation was nearly perfect and may encourage practitioners to implement its use baring all future comparisons and the long term monitoring of their athlete continue to be derived from the smart phone application.

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FIGURE CAPTIONS

Figure 1. Relationship between $rMSSD_{ln}$ derived from Elite HRV smart phone app and Kubios HRV 2.2 (r = 0.92; CI 95% = 0.90-0.93; $p < 0.0001$)

Figure 2. Bland-Altman Plot comparing % residuals in $rMSSD_{ln}$ derived from Elite HRV and Kubios HRV 2.2. The solid line represents the mean bias (-2.7%; CI 95% = -3.1 to -2.3%). The two dashed lines outside of the mean represent 95% levels of agreement (1.96SD; 7% to -12%). The dashed line in the middle represents the line of equality (0%)

Figure 3. Q-Q plot comparing the theoretical rMSSD quintiles to the observed rMSSD quintiles derived from Elite HRV.

Figure 4. Q-Q plot comparing the theoretical $rMSSD_{ln}$ quintiles to the observed $rMSSD_{ln}$ quintiles derived from Elite HRV.

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Table 1. Descriptive statistics (mean ± SD) of the participants studied.

		Mean ± SD
Age (9)		30.8 ± 11.7
Gender	Male = 29	
	Female = 8	
Height (cm)		177.9 ± 9.6
Weight (kg)		85.1 ± 14.3
BMI (kg-m²)		26.8 ± 4.0

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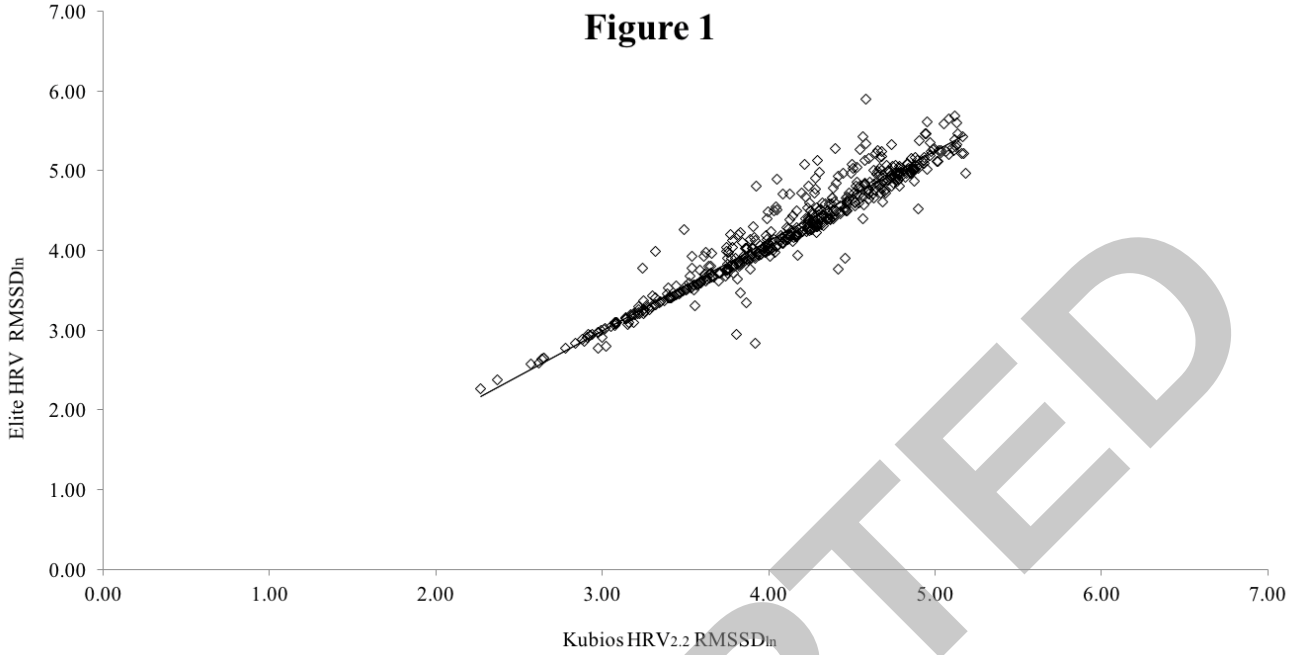

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Figure 2

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Figure 3

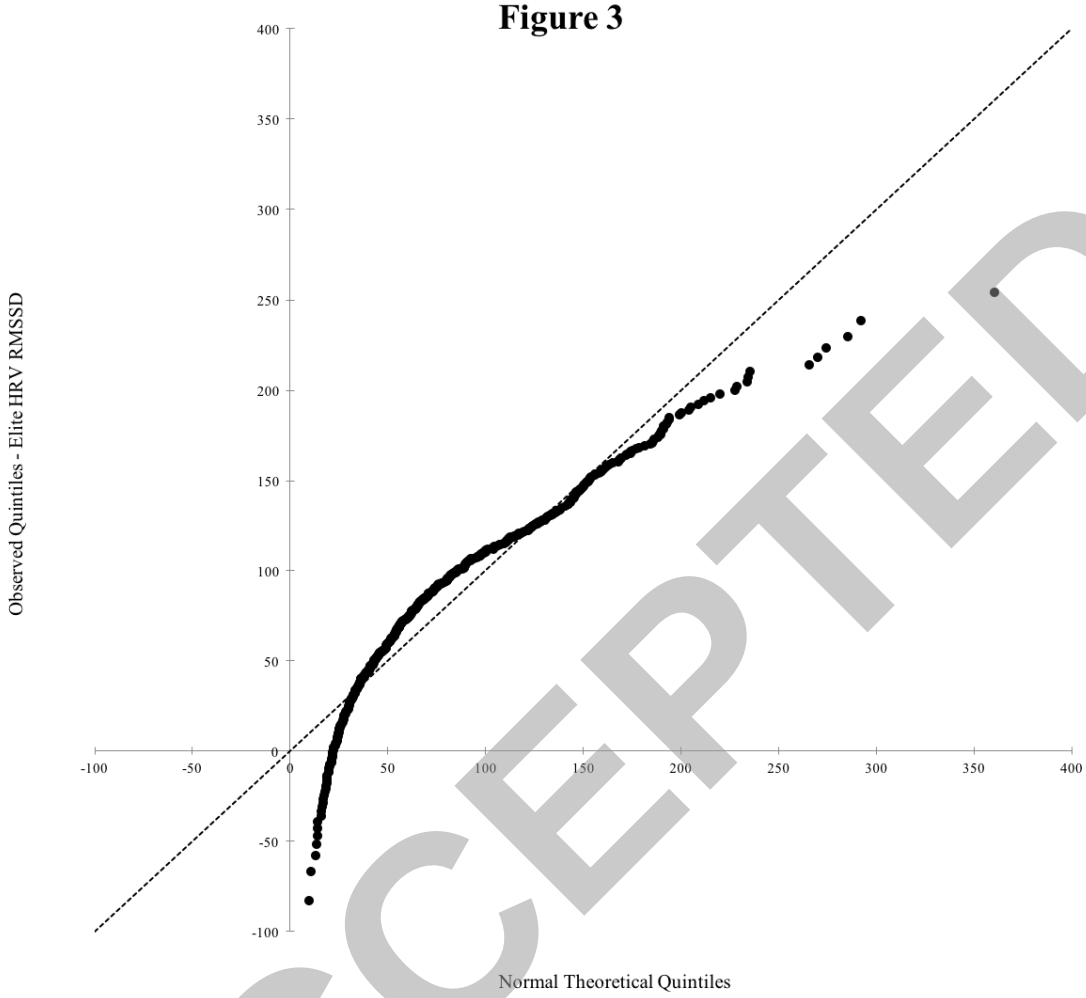

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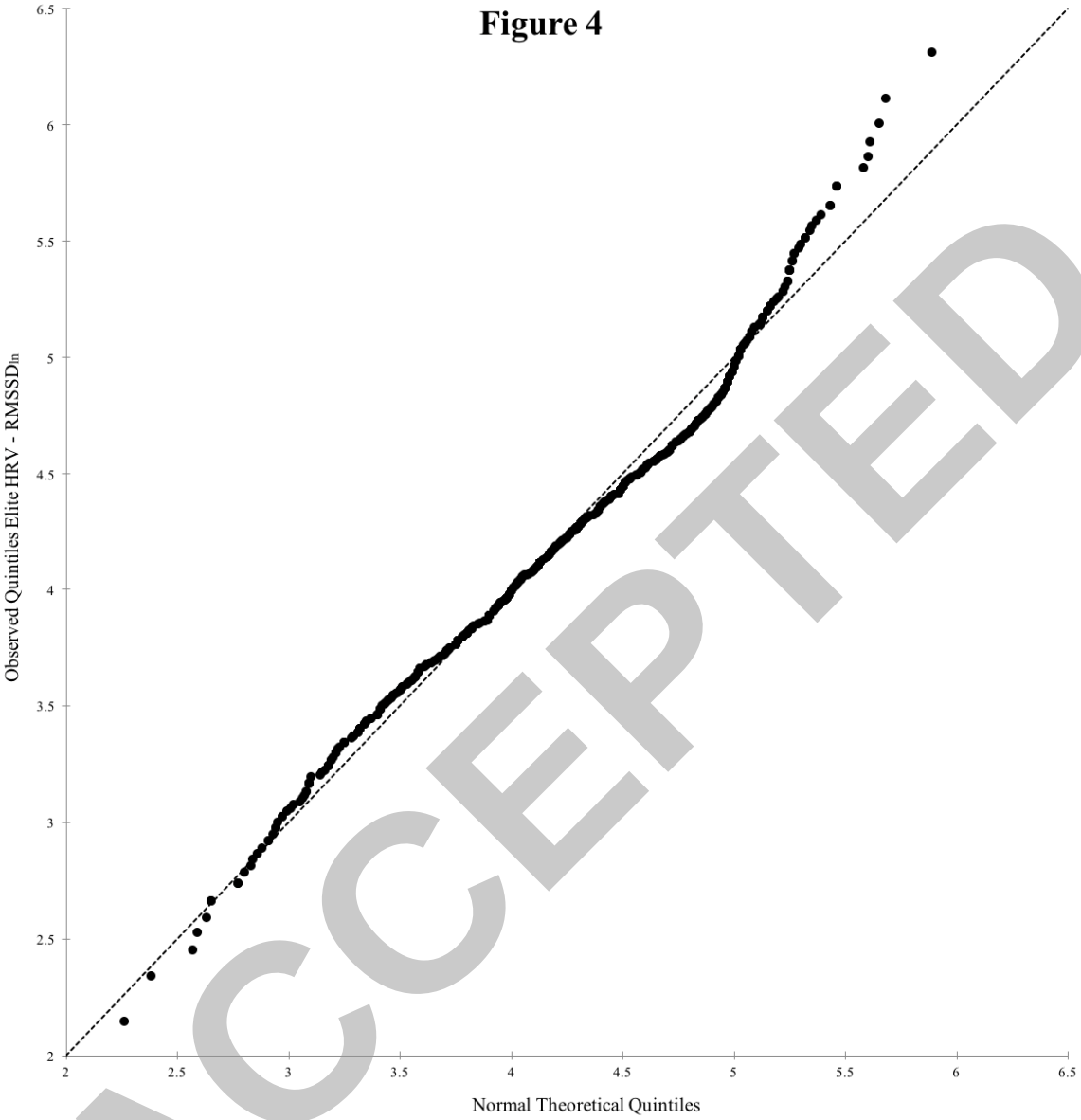
Figure 4



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